# EV Battery Course — Module 12: Lithium-Ion Battery Manufacturing Process

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A cell factory is three plants in one: a precision coating plant (electrodes), a clean assembly plant (cells), and a huge charging warehouse (formation and ageing). Then a pack plant turns cells into a vehicle part. This module walks all 25 steps with equipment, critical parameters, defects, checks, automation and EHS, then covers dry rooms, solvent recovery, energy, layout, cost and Industry 4.0.

## 1. Learning objectives

- Describe each of the 15 cell-manufacturing and 10 pack-manufacturing steps.
- Identify critical process parameters, typical defects and quality checks for each.
- Explain dry-room requirements, solvent recovery and formation energy.
- Estimate line throughput, solvent use, WIP and yield.
- Identify where robotics, vision, digital twins and predictive maintenance pay off.

## 2. Fundamentals: the three cell sections

| Section | Steps | Environment | Share of cost / time |
|---|---|---|---|
| Electrode manufacturing ("front end") | Mixing → coating → drying → calendering → slitting | Clean, controlled humidity; NMP solvent handling | Largest equipment and energy share |
| Cell assembly ("middle") | Notching → stacking/winding → tab welding → packaging → filling | Dry room (dew point about −40 °C or drier) | Precision and cleanliness critical |
| Cell finishing ("back end") | Wetting → formation → degassing → ageing → grading | Temperature-controlled racks; fire safety | Long time, big WIP, significant energy |

# 3. Process flow diagram

```
flowchart TB
  subgraph E["Electrode manufacturing"]
    M["1. Raw material prep"] --> SL["2. Slurry mixing"] --> CO["3. Coating"]
--> DR["4. Drying + NMP recovery"] --> CA["5. Calendering"] --> SN["6.
Slitting and notching"]
  end
  subgraph A["Cell assembly (dry room)"]
    SW["7. Stacking or winding"] --> TW["8. Tab welding"] --> AS["9. Assembly
and sealing"] --> EF["10. Electrolyte filling"]
  end
  subgraph F["Cell finishing"]
    WE["11. Wetting"] --> FO["12. Formation"] --> DG["13. Degassing"] -->
AG["14. Ageing"] --> GR["15. Grading and inspection"]
  end
  subgraph P["Pack plant"]
    P1["Incoming cell inspection"] --> P2["Matching"] --> P3["Module/pack
assembly"] --> P4["Busbar welding"] --> P5["Cooling integration"] --> P6["BMS
and sensors"] --> P7["Enclosure sealing"] --> P8["Electrical safety tests"] -
-> P9["EOL functional test"] --> P10["Traceability and dispatch"]
  end
  SN --> SW
  EF --> WE
  GR --> P1
```

# 4. Cell manufacturing, step by step

| # | Step: purpose and equipment | Critical process parameters | Common defects | Quality checks |
|---|---|---|---|---|
| 1 | **Raw material prep:** dry and dose active material, conductive carbon, binder; vacuum dryers, gravimetric feeders | Moisture, dosing accuracy, foreign matter | Wet powder, wrong ratio, metal particles | KF moisture, PSD, magnetic separation, lot release |

| # | Step: purpose and equipment | Critical process parameters | Common defects | Quality checks |
|---|---|---|---|---|
| 2 | **Slurry mixing:** disperse into NMP (cathode) or water (anode); planetary or continuous twin-screw mixers | Mixing energy, time, temperature, vacuum, solid content | Agglomerates, bubbles, sedimentation | Viscosity, solid content, filtration, Hegman gauge |
| 3 | **Coating:** slurry onto foil; slot-die coaters, often double-sided, 50–100 m/min | Gap, pump flow, web speed and tension, edge profile | Thickness variation, streaks, pinholes, edge buildup | Online beta/X-ray weight, vision |
| 4 | **Drying:** evaporate solvent; multi-zone convection ovens up to ~100 m long | Zone temperatures, airflow, speed; NMP concentration | Cracking, binder migration, residual solvent | Residual solvent, adhesion, visual |
| 5 | **Calendering:** compress to target density; heated roll press | Roll gap, force, temperature, speed | Over-densification (poor wetting), wrinkles, foil breaks | Thickness gauge, density, surface vision |
| 6 | **Slitting and notching:** cut to width and form tabs; rotary knives, laser notching | Blade condition, laser power, speed | Burrs, dust, heat-affected zones | Burr microscopy, dust extraction monitoring |

| # | Step: purpose and equipment | Critical process parameters | Common defects | Quality checks |
|---|---|---|---|---|
| 7 | **Stacking or winding:** build electrode assembly; Z-fold stackers or winders | Alignment, tension, speed | Misalignment, separator folds, overhang loss | CCD vision, X-ray, hi-pot |
| 8 | **Tab welding:** join foil tabs to terminals; ultrasonic or laser | Energy, pressure, time, cleanliness | Weak welds, foil tears, spatter | Pull test, resistance, vision |
| 9 | **Assembly and sealing:** insert into can/pouch; laser seam weld or heat seal | Weld power/speed, seal temperature/pressure/time | Pinholes, cracks, poor seals | Helium leak test, hi-pot, X-ray |
| 10 | **Electrolyte filling:** vacuum dosing in dry environment | Dose mass, vacuum level, dew point | Under/overfill, spills, moisture uptake | Weighing before/after |
| 11 | **Wetting:** rest so electrolyte soaks pores; sometimes heat and pressure cycling | Time, temperature | Dry spots → plating | Process time control |

| # | Step: purpose and equipment | Critical process parameters | Common defects | Quality checks |
|---|---|---|---|---|
| 12 | **Formation:** first slow charges forming the SEI; cyclers with energy recovery, often at elevated temperature and with pressure for pouches | Current profile, temperature, pressure, time | Abnormal voltage, gas, early shorts | Voltage curve analytics, capacity |
| 13 | **Degassing:** remove formation gas (pouch and some prismatic) and reseal; vacuum stations | Vacuum, seal parameters | Residual gas, electrolyte loss | Thickness, weight |
| 14 | **Ageing:** rest days to weeks at room/high temperature | Time, temperature | Undetected micro-shorts escape if too short | OCV/K-value self-discharge screening |
| 15 | **Grading and final inspection:** capacity, DCIR, OCV, dimensions, appearance; sorting into bins | Measurement accuracy, temperature | Mis-binning | MSA, SPC on capacity/resistance distributions |

## Automation and EHS by section

- *Electrode:* high automation already; additions are closed-loop coating-weight control, AI vision defect classification, predictive maintenance on rolls and pumps. EHS: NMP exposure (reproductive toxicant) and solvent fire risk; dust explosion controls; noise.
- *Assembly:* robotic handling, vision-guided alignment, high-speed laser welding. EHS: laser safety, dry-room working conditions (dehydration), electrolyte handling (HF risk).
- *Finishing:* automated storage and retrieval systems (AS/RS) for formation and ageing racks; regenerative cyclers return energy to the grid. EHS: thousands of charged cells → fire detection, sprinklers or isolation, thermal camera monitoring, emergency removal of hot cells.

**Productivity and yield metrics:** OEE (overall equipment effectiveness), first-pass yield per step, scrap rate ($m^2$ of electrode, cells), cycle time (ppm – parts per minute), changeover time, energy per kWh, and kWh output per employee.

## 5. Pack manufacturing, step by step

| # | Step | Key parameters / checks | Typical defects |
|---|---|---|---|
| 1 | Incoming cell inspection and grading | OCV, ACIR, dimensions, visual, data from cell supplier | Transport damage, out-of-spec OCV |
| 2 | Cell matching and grouping | Capacity and resistance bins; OCV window | Mixed bins → imbalance |
| 3 | Module or pack assembly (stacking, gluing, compression) | Adhesive bead, cure, compression force, insulation films | Misplaced cells, insufficient insulation, adhesive voids |
| 4 | Busbar welding (laser, ultrasonic, wire bonding for cylindrical) | Weld power, focus, cleanliness; joint resistance | Undersized welds, spatter, cell damage from heat |
| 5 | Cooling system integration | TIM bead weight/coverage, plate fit, hose connections | TIM voids, coolant leaks |
| 6 | BMS and sensor installation | Harness routing, connector seating, thermistor placement | Pinched wires, wrong channel mapping |
| 7 | Enclosure sealing | Gasket or sealant bead, bolt torque sequence | Water ingress paths |

| # | Step | Key parameters / checks | Typical defects |
|---|---|---|---|
| 8 | Electrical safety testing | Insulation resistance, dielectric withstand, potential equalisation (chassis bonding) | Isolation faults |
| 9 | EOL functional test | BMS comms, all cell voltages and temperatures, contactors, pre-charge, HVIL, current-sensor calibration, short charge/discharge | Software/configuration errors, sensor faults |
| 10 | Traceability and dispatch | Pack ID linked to every cell ID, test data, SOC set for shipping (UN 38.3 / dangerous goods) | Missing records, wrong SOC |

## 6. Dry rooms, solvent recovery, energy, layout and cost

**Dry rooms.** Electrolyte salt $LiPF_6$ reacts with water to form HF; moisture also degrades high-nickel cathodes. Assembly is done in dry rooms at dew points around −40 °C to −60 °C. Desiccant-wheel dehumidifiers run continuously; people are the biggest moisture source (breath, skin), so access is restricted and airlocks are used. Dry rooms are a major energy and capex item.

**Solvent recovery.** NMP used for cathode slurries is expensive and toxic; dryer exhaust passes through condensers and/or adsorption to recover about 99%+, which is distilled and reused. Water-based cathode processing and dry-electrode processes (Module 16) aim to eliminate NMP.

**Formation energy.** Cells are charged and discharged in formation and grading; modern regenerative cyclers recover much of the discharge energy. Studies estimate total cell-plant energy demand on the order of 20–40 kWh per kWh of cell capacity for today's plants, with drying and dry rooms the largest consumers (see Degen et al., *Nature Energy* 2023).

**Factory layout.** Linear flow from raw material warehouse through electrode, assembly (inside dry-room envelope) and finishing (large rack halls), with fire separation between formation/ageing and the rest. A 10 GWh/year plant typically occupies tens of thousands of $m^2$.

**Cost drivers (cell):** materials (about 60–75% of cell cost: cathode largest), depreciation of equipment and buildings, energy, labour, yield losses and scrap, and overhead. Yield is the hidden lever: scrap late in the process wastes almost all material value.

## 7. Engineering equations

$$\text{Areal capacity (Ah/m}^2) = \text{loading (g/m}^2) \times f_{active} \times q\ (\text{Ah/g})$$

$$\text{Line output (kWh/h)} = v\ (\text{m/min}) \times w\ (\text{m}) \times 60 \times \frac{\text{areal energy (Wh/m}^2)}{1000}$$

$$Y_{total} = \prod_i Y_i \qquad OEE = \text{Availability} \times \text{Performance} \times \text{Quality}$$

$$\text{WIP (kWh)} = \text{daily output} \times \text{ageing days}$$

## 8. Worked examples

**Example 1: coater throughput.** LFP cathode, loading 200 g/m² per side, 96% active, 155 mAh/g, double-sided, 3.2 V; line speed 60 m/min, usable width 1.2 m.

- Areal capacity = 200 × 0.96 × 0.155 = 29.8 Ah/m² per side → 59.5 Ah/m² both sides → 190 Wh/m².
- Area = 60 × 1.2 = 72 m²/min → 72 × 0.19 = **13.7 kWh/min = 823 kWh/h**.
- At 6,000 productive hours/year: **~4.9 GWh/year** of cathode from one coater (before scrap).

**Example 2: NMP use.** Slurry 70% solids → 0.43 kg NMP per kg solids. LFP needs about 2.1 kg cathode coating solids per kWh → **0.90 kg NMP/kWh** → for 10 GWh: **~9,000 t/year** of NMP circulating. At 99% recovery, about 90 t/year must be replaced, and the rest is recycled.

**Example 3: how dry is a dry room.** At −40 °C dew point, water vapour ≈ 0.095 g/m³. Typical Indian factory air at 25 °C and 60% RH holds about 13.8 g/m³: **~145× drier**.

**Example 4: cumulative yield.** Coating 99.5% × calendering/slitting 99.0% × assembly 99.0% × formation/ageing/grading 98.5% = **96.1%**. About 4% of material value is lost, much of it after most value has been added.

**Example 5: ageing WIP.** A 10 GWh/year plant makes 27.4 MWh/day; 14 days of ageing → **384 MWh** of cells on racks, about **750,000 cells** of 512 Wh. That is working capital and a fire-safety design case.

## 9. Industry 4.0 opportunities

| Technology | Where | Benefit |
|---|---|---|
| Robotics | Cell handling, module stacking, busbar welding, TIM dispensing | Speed, consistency, handling in dry rooms |
| Machine vision and AI | Coating defects, alignment, weld seams, adhesive beads | Higher detection, fewer false rejects |
| Digital twins | Coating and drying process models; factory flow simulation | Faster ramp-up, fewer trial runs |
| Predictive maintenance | Coater pumps, rollers, laser optics, dehumidifiers | Less downtime and contamination |
| Data traceability and analytics | Cell genealogy, formation curve analytics | Earlier defect prediction; targeted recalls |
| Energy management | Regenerative formation, heat recovery from drying | Lower cost and $CO_2$ |

Ramp-up is the hardest phase of a gigafactory: many new plants have struggled for years to reach target yields (scrap rates of 10–30% early in ramp-up are commonly cited in industry commentary), which is why process expertise is valued so highly.

## 10. Real-world examples

- **CATL, BYD, LGES, Panasonic, Samsung SDI, SK On:** the leading global cell makers, with decades of process know-how.
- **Northvolt (Sweden):** filed for bankruptcy in 2024–25 after severe ramp-up and yield problems, a widely discussed lesson on how hard scaling is.
- **India:** Ola Electric's cell plant in Tamil Nadu, Tata Agratas (Sanand, Gujarat), Amara Raja and Exide Energy cell plants are at various stages of construction and ramp-up; see Module 15 for current status.

## 11. Common mistakes and misconceptions

- **"Cell making is like assembly."** It is closer to chemical and precision coating processes.

- **"Equipment buys capability."** Process recipes, materials know-how and people make the yield.
- **"Formation is just charging."** Formation recipes strongly influence life and safety.
- **"Dry rooms are optional in humid climates."** They matter more in humid climates such as India's.

## 12. Practical case study: coating streaks after a slurry change

**Problem.** After switching to a new conductive carbon supplier, coating vision rejects rise from 0.5% to 4%, mostly streaks.

**Investigation.** New carbon has higher surface area, increasing slurry viscosity and forming agglomerates that lodge in the slot-die lip.

**Action.** Adjust mixing energy and dispersant level; add inline filtration; update incoming spec for carbon surface area; requalify per change-management (PPAP) process.

**Lesson.** Small material changes ripple through the front end; change control is essential.

## 13. Revision notes

- 15 cell steps in three sections; 10 pack steps.
- Electrode quality (weight, burrs, contamination) drives everything downstream.
- Dry rooms (−40 °C dew point or below), NMP recovery (~99%), and formation/ageing are big cost and energy items.
- Yield is multiplicative; late-stage scrap is most costly.
- Ramp-up know-how is the true barrier to entry.

## 14. Glossary

| Term | Meaning |
|---|---|
| Slot-die | Precision coating head that meters slurry onto foil |
| Calendering | Roll pressing of coated electrodes |
| NMP | N-methyl-2-pyrrolidone, cathode slurry solvent |
| Dew point | Temperature at which moisture condenses; measure of dryness |
| Formation | Initial controlled cycling that forms the SEI |

| Term | Meaning |
|---|---|
| OEE | Overall equipment effectiveness |
| AS/RS | Automated storage and retrieval system |
| WIP | Work in progress |

## 15. Knowledge check

### Conceptual

1. Why is cell assembly done in a dry room but coating often is not?
2. Why does overcalendering hurt performance?
3. Why are formation and ageing major cost drivers?
4. What is the purpose of degassing?
5. Why is ramp-up so difficult for new gigafactories?

### Numerical

1. A coater runs 40 m/min at 1.0 m width producing 180 Wh/m². Annual output at 6,500 h?
2. Step yields 99.8%, 99.2%, 98.8%, 99.0%, 98.0%. Total yield?
3. A 5 GWh/year plant ages cells for 10 days. WIP in MWh?

### Application

1. As a cost engineer, how would you build a should-cost model for a 100 Ah LFP cell?
2. An Indian plant suffers monsoon-season defects. Which processes and controls would you check?

## 16. Answer key

### Conceptual

1. Once electrolyte or moisture-sensitive cell internals are exposed (assembly and filling), water reacts with $LiPF_6$ to make HF. Electrodes are dried later and protected; coating happens with solvent present.
2. Pores close, so electrolyte can't wet the electrode well; ion transport slows, causing poor rate performance and plating risk; particles may crack.
3. Days to weeks of rack time per cell: large buildings, many cyclers, energy, WIP inventory and fire protection.
4. To remove gas produced during formation that would otherwise cause swelling, poor contact and plating.

5. Many interacting process parameters, new equipment, materials variability, and staff learning; yields and OEE start low and improve slowly.

**Numerical**

1. 40 × 1.0 × 60 = 2,400 m²/h × 0.18 kWh = 432 kWh/h × 6,500 = **2.81 GWh/year**.
2. 0.998 × 0.992 × 0.988 × 0.990 × 0.980 = **0.9488 (94.9%)**.
3. 5,000 MWh ÷ 365 = 13.7 MWh/day × 10 = **137 MWh**.

**Application**

1. Materials: kg per cell of each material × price (from Module 3: cathode, anode, electrolyte, separator, foils, can, lid) plus scrap. Conversion: per-step machine hour rates (depreciation, energy, maintenance) × cycle time, labour, dry-room and formation energy, overhead. Divide by yield. Compare with market cell prices (BNEF 2025).
2. Dry-room dew point logs and dehumidifier capacity; airlock and gowning discipline; moisture of incoming materials and stored electrodes; electrode re-drying times; electrolyte storage; correlation of defects (gas, K-value) with humidity.

## 17. Further learning

- Kwade et al., "Current status and challenges for automotive battery production technologies," *Nature Energy* 2018.
- Degen et al., "Energy consumption of current and future production of lithium-ion and post lithium-ion battery cells," *Nature Energy* 2023.
- PEM RWTH Aachen and VDMA, *Lithium-ion Battery Cell Production Process* brochures (free).
- Heimes et al., *Battery Production* roadmaps (PEM RWTH Aachen).

Next: Module 13, safety, failure analysis and functional safety.